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Collaborative sorting: integrating pneumatic separation and manipulator techniques for efficient coal gangue sorting

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ABSTRACT

Coal is one of the important sources to meet the global energy demand. The intelligent treatment of coal gangue is of great significance to environmental protection and resource recycling. In the treatment of gangue, the use of manipulator sorting alone faces the challenge of being difficult to adapt to the increasing sorting intensity and high cost of use. In this study, we propose a hybrid sorting system that combines pneumatic separation and manipulator sorting to address the limitations of pneumatic separation sorting and manipulator. We design a task assignment algorithm based on genetic algorithms to realize their collaborative work for optimal overall sorting efficiency in a sequence of gangue. Experimental results demonstrate that the collaborative sorting system, incorporating both manipulator and high-pressure pneumatic separation, outperforms the single pneumatic separation sorting system with an average increase in sorting efficiency of 20.66%. Furthermore, the collaborative system increased sorting efficiency by up to 76.09% compared to single manipulator sorting. These findings validate the effectiveness of our proposed approach in achieving efficient sorting of coal gangue.

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Pneumatic separation;
manipulator; collaborative
sorting; genetic algorithms

Introduction

Gangue is a solid waste produced in the process of mining production, which usually contains a large number of useless or harmful substances, such as heavy metals, organic matter, radioactive substances, etc. Therefore, the separation of coal and gangue is an indispensable part of the coal production process (Fan, Zhang, and Wang 2014; Yang et al. 2019). Traditional gangue separation methods such as heavy media method (Meyer and Craig 2015), jigging method (Ambrós 2020) and flotation method (Zhou et al. 2020), mechanical vibration method (Yang and Zeng 2021) are gradually eliminated due to the problems of easy loss of separation equipment, high energy consumption, high cost of drying treatment and media recovery (Zhao et al. 2014; Zou et al. 2020).

In recent years, intelligent recognition technology for coal gangue has driven the development of sorting technology toward intelligence, automation, and efficiency (Jiang et al. 2023; Lv et al. 2022; Xin and Dong 2022). Liu et al. (2019) proposed a motion model for coal gangue picking robots based on a machine vision system. Sun, Huang, and Jia (2021) improved the six-axis manipulator system to achieve efficient motion planning. Despite

the highly automated and accurate nature of manipulator sorting, the reciprocating motion consumes a significant amount of time, thereby affecting the overall efficiency of the sorting process. To address this issue, researchers propose multi-manipulator system to enhance the working capacity of the sorting system (J. Li, Dong, and Li 2016; S. D. Li, Xu, and Zuo 2015; Sarel and Balch 2006; Shang et al. 2020; Sharma et al. 2014; Wang et al. 2023; Xiangang et al. 2019). With the increase in coal mining volume and intensity, higher requirements have been placed on the belt speed and sorting efficiency of coal gangue sorting lines. To meet these demands, it is necessary to adopt more manipulators and more efficient collaborative working schemes. However, multi-manipulator systems not only pose higher control difficulties, but also challenges such as increased cost, complexity, and failure rates of the manipulator system.

Compared with manipulator, pneumatic separation usually has the advantages of lower equipment cost and simple control system, which are widely used in solid waste treatment and ore separation. Bi et al. (2019) proposed a method to recover copper and aluminum fragments from the crushed products of waste lithium iron phosphate batteries according to aerodynamic principles. De Almeida et al. (2019) developed a pneumatic classification model for bagasse particles. Zheng et al. (2015; 2016) derived a theoretical formula for the gangue pneumatic separation distance and also proposed two different pneumatic separation schemes, which elucidated the important factors affecting the gangue pneumatic separation distance. However, the gangue is treated as a perfect sphere in the aforementioned studies, yet in reality, its types and morphology vary significantly. This makes it challenging for a single separation system to accommodate diverse gangue characteristics, often resulting in incorrect or missed picking. Consequently, the separated gangue require further screening, washing, or manual sorting, ultimately adding complexity to the process and increasing labor costs.

In this study, we propose a coal gangue online sorting model that combines pneumatic separation as the base sorting method with manipulator as a supplementary sorting method. By analyzing the sorting reliability of pneumatic separation for gangue of different qualities and morphologies, we determine the sorting priority of mechanical arms for different types of gangue. To fully utilize the sorting capability of mechanical arms and avoid task conflicts or overlooking higher-value targets during the sorting process, we consider a certain length of gangue sequence as the basic unit and use a genetic algorithm to optimize the operation objectives of the manipulators. The objective is to maximize the sorting efficiency by allocating tasks in a rational manner. To validate the feasibility and effectiveness of this approach, we conducted sorting experiments under different belt speeds and gangue content levels. These research findings are of significant importance for the development of coal gangue sorting technology and improving the efficiency and accuracy of coal gangue sorting processes.

Methods

Composition of Collaborative Sorting System

The coal gangue collaborative sorting system is designed to efficiently separate gangue and coal blocks based on their characteristics. As shown in Figure 1, It consists of an identification system, a control system and an actuation system. The

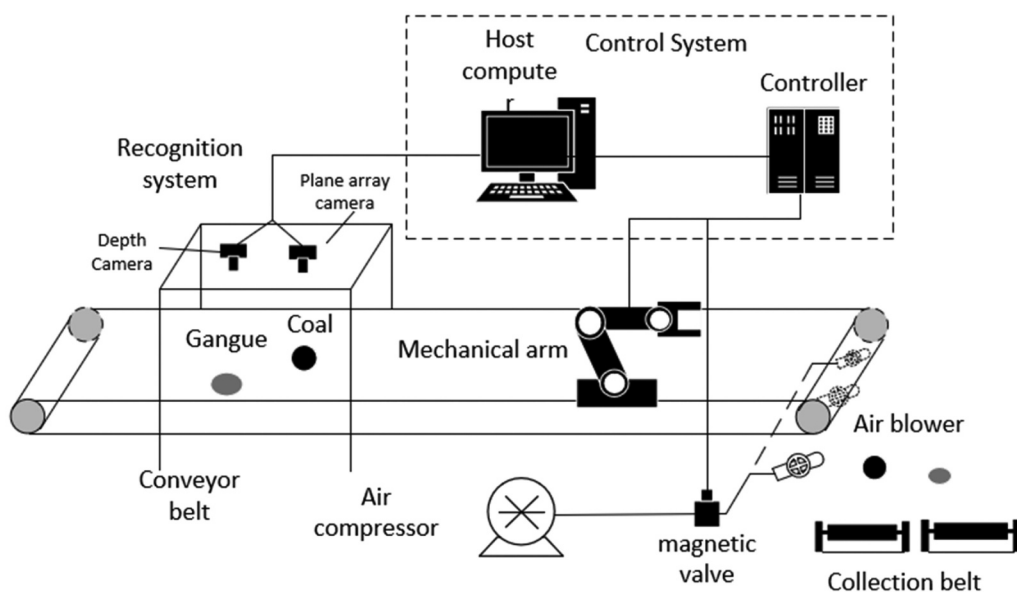
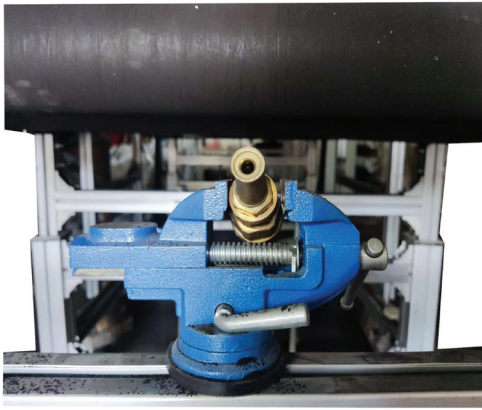


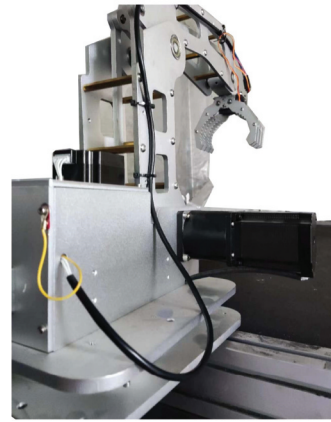
Figure 1. Coal gangue cooperative sorting system composition.

vision section comprises an area scan camera and a depth camera. The area scan camera, Hikvision MV-CS050-10UC, is used to capture RGB images of gangue and coal blocks on the conveyor belt. These images are utilized for classification and localization of the gangue and coal blocks. The Kinectv2 depth camera is employed to acquire point cloud information of coal gangue, extracting three-dimensional characteristics of individual gangue particles. This information is used to estimate the quality of gangue particles, which serves as a basis for subsequent gangue sorting control.

Control system including host computer and PLC controller. The host computer, with image recognition, point cloud processing, target localization and collaborative sorting algorithms, plays a crucial role in coordinating the sorting process. By employing collaborative sorting strategies, the upper computer allocates tasks efficiently to the actuators. The sorting execution system consists of two major components: pneumatic sorting and mechanical sorting mechanisms. The pneumatic sorting mechanism is shown in Figure 2(a), in order to ensure the actuator force, the power of the pressurizer is 7.5 kw, and the maximum pressure is 1.25 Mpa and three nozzles with a diameter of 12 mm are selected. In this paper, the SMCVT307 solenoid valve is used, and the maximum frequency can be up to 10 times/s, which can also meet the speed requirements of sorting when the amount of coal is large. Mechanical sorting mechanism shown in Figure 2(b), the mechanical arm for the four-axis mechanical arm, 680 mm arm span, can load 0.8 ~ 2.5 kg of objects, the maximum operating speed of up to 480 °/s, the maximum power of 500 W can meet the majority of the test gangue.



(a)



(b)

Figure 2. (a) Pneumatic separation mechanism (b) Manipulator.

Manipulator and Pneumatic Separation Coordination Sorting Principle

The core of collaborative sorting includes the reliability of air-blow sorting for different types of gangue, feasibility assessment of mechanical hand sorting tasks, and rational task allocation strategies. Pneumatic separation's reliability for various gangue types is used to evaluate how gangue is assigned. Based on this reliability, gangue is assigned to either the pneumatic separation mechanism or manipulator. The variations in reliability indicate the priority of assigning targets to the air-blow mechanism or the manipulator. The purpose of the task allocation strategy is to enable the manipulator and air blow to flexibly select sorting targets from a large number of gangue to achieve the best overall sorting efficiency of the gangue sequence, rather than simply selecting high priority targets.

The collaborative sorting process consists of three steps. First, the position, quality and shape information of the gangue, as well as the state information of the manipulator and the pneumatic mechanism are obtained. The current gangue information is sorted into a task sequence based on the x-coordinate and inputted into the task allocation model. The task allocation algorithm assigns task targets to each executor based on the current gangue sequence and conducts feasibility analysis to eliminate task conflicts. Then, the sorting efficiency of the task allocation scheme is calculated. This process is iterative in order to find the best task allocation scheme. Finally, the feasible allocation results are outputted to the executors and the next gangue sequence is updated. The collaborative sorting process is illustrated in Figure 3.

Collaborative Sorting Task Assignment Strategy

Reliability Analysis of Pneumatic Separation

In the Pneumatic separation method, the trajectory of gangue particles is influenced by various factors such as airflow pressure, flow rate, direction, incidence angle, and belt running speed. However, these factors are general influences of the sorting equipment on

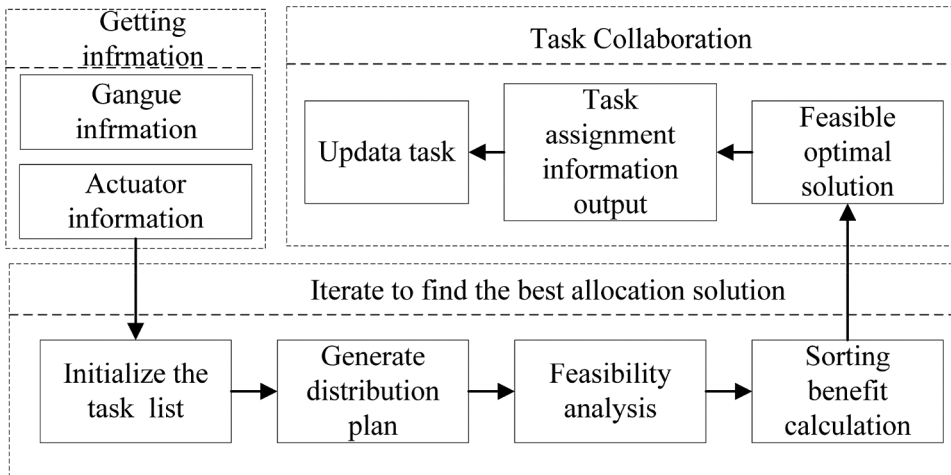


Figure 3. Collaborative sorting process.

gangue sorting and cannot be used as individual criteria for task allocation in the sorting process. The distinguishing factors among gangue particles lie in their quality and form, requiring an analysis of how gangue quality and shape impact the separation distance during sorting.

The effectiveness of pneumatic separation on gangue sorting is affected by the different sizes of gangue particles, which exhibit varying levels of resistance and reaction to airflow. Generally, larger gangue particles have greater weight and inertia, resulting in a weaker response to airflow and a lower probability of successful sorting. Conversely, smaller gangue particles, due to their lighter and more flexible nature, exhibit a more pronounced response to airflow, leading to a higher probability of successful sorting. Additionally, the different shapes of gangue particles result in varying force areas within the airflow domain, further contributing to differences in sorting outcomes. The classification of gangue shapes can be determined by the ratio between the minimum length of the bottom surface of the gangue and the height of the gangue. Sampling of the laboratory gangue, [Figure 4](#) shows the classification thresholds, mass distribution and proportion of gangue. [Figure 5](#) illustrates the varied shapes of gangue within the airflow domain, with the primary force acting on the lower surface of the gangue.

To evaluate the sorting reliability of the pneumatic separation method, a pneumatic separation model was developed, as depicted in [Figure 6\(a\)](#). The parameters that appear below include h_0 (distance from the nozzle before the gangue particles fall), h_1 (height of fall of gangue after force), θ (rotation angle of the pneumatic separation device), l (distance of action of air blowing), x (total horizontal movement distance of the gangue particle), x_0 (horizontal movement distance of the gangue particle before entering the airflow domain), and x_q (horizontal movement distance of the gangue particle after airflow domain), m (mass of the gangue particle).

To facilitate the analysis of gangue movement in the airflow area, we made the following assumptions:

- (1) Neglect the air resistance effect on the falling gangue;

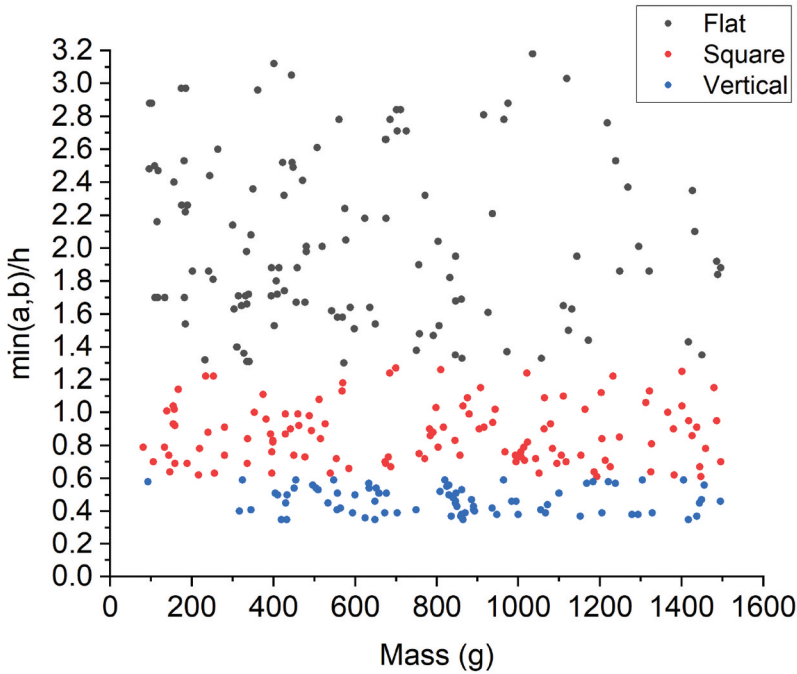


Figure 4. Classification thresholds, mass distribution and proportions of gangue.

- (2) Consider the action time of high-pressure gas to be extremely short, on the millisecond scale, during which the gangue's movement distance is not taken into account;
- (3) Assume the gas pressure to be a constant force during the action time of high-pressure gas.

The displacement distance of the gangue before entering the airflow domain can be calculated using Eq. (1), where v_0 is belt speed, t_0 is the movement time of the gangue before it enters the airflow-domain.

$$\begin{cases} l \cos \theta = v_0 t_0 \\ \frac{1}{2} g t_0^2 + l \sin \theta = h_0 \\ x_0 = v_0 t_0 \end{cases} \quad (1)$$

When gangue falls into the airflow domain, it experiences both gravity and the impact of gas pressure. The forces acting on the gangue in the airflow domain are illustrated in [Figure 6\(b\)](#). The force exerted by high-pressure gas on the gangue, denoted as F_q , can be determined using Eq. (2). Prior to the high-pressure gas action, the y-direction velocity (v_y) of the gangue can be calculated using Eq. (3). Where A is the gangue force area, p is the airflow pressure.

$$F_q = \int p dA = \frac{1}{2} \rho v^2 A \quad (2)$$

$$v_y = \sqrt{2gh_0} \quad (3)$$

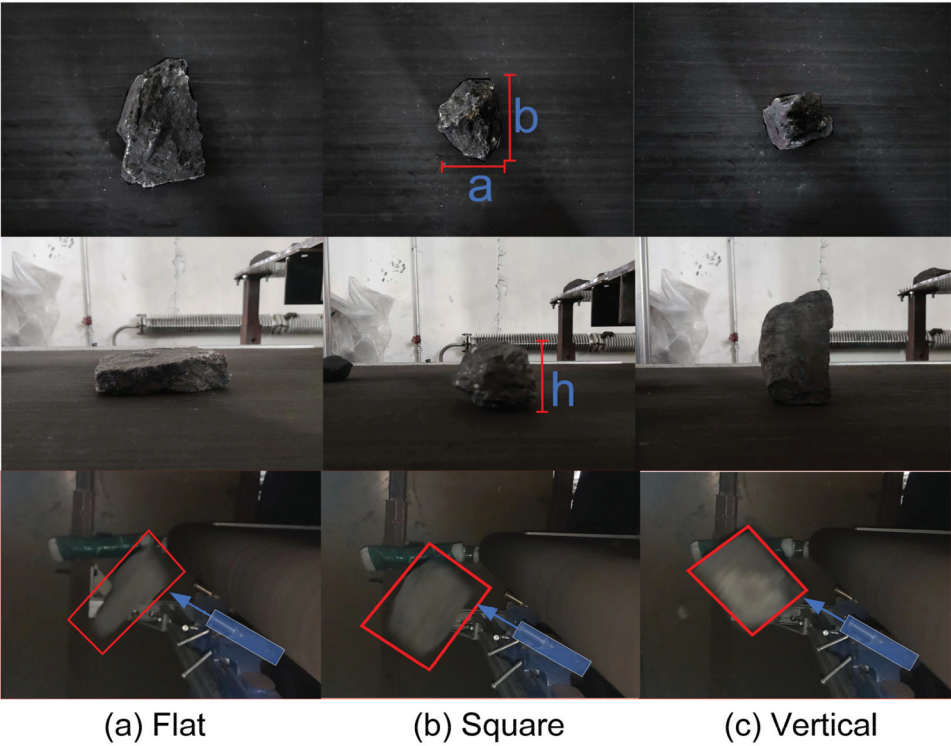


Figure 5. Different shapes of gangue and force surfaces.

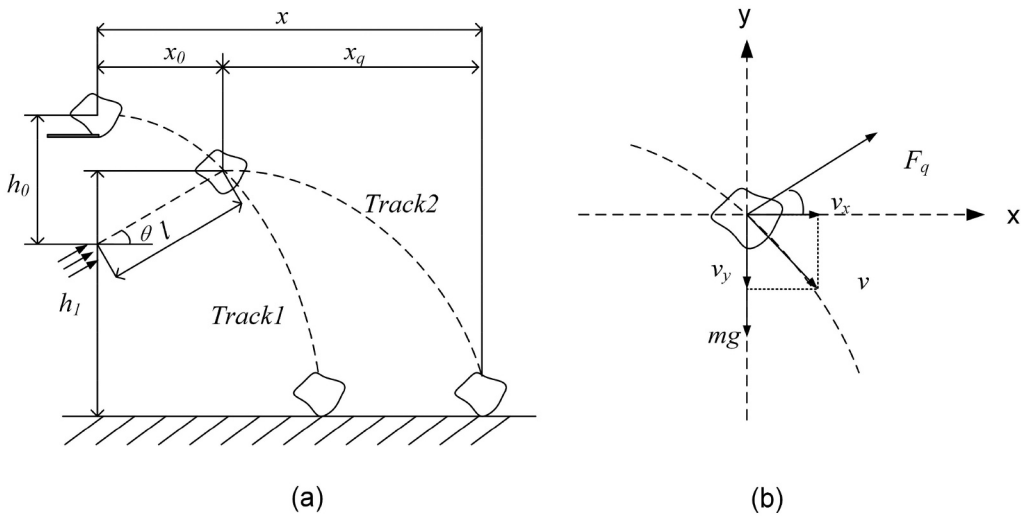


Figure 6. (a) is the gangue movement trajectory diagram. (b) is the force diagram of the gangue in the airflow domain.

After the high-pressure gas force ceases, the gangue's movement trajectory undergoes a change. At this stage, the horizontal and vertical velocities (v_{x1} , v_{y1}) of the gangue can be obtained from Eq. (4) and Eq. (5), respectively, where t_q is duration of air flow action.

$$v_{x1} = v_0 + \int_0^{t_q} a_x dt = v_0 + \frac{F_q t_q \cos \theta}{m} \quad (4)$$

$$v_{y1} = v_y + \int_0^{t_q} a_y dt = v_y + \frac{(F_q \sin \theta - mg)t_q}{m} \quad (5)$$

After the high-pressure airflow stops, the gangue continues to carry out horizontal throwing motion. At this stage, the movement trajectory of the gangue is shown in Figure 6(a) Track2. According to the vertical falling height of the gangue, the movement time of the gangue after air blowing (t_d) can be calculated, followed by the displacement of the gangue in the horizontal direction after air blowing (x_q), and finally the displacement before and after the air blowing is added to derive the total displacement (x), which is calculated by Eq. (6).

$$\begin{cases} x_q = v_{x1} t_d \\ h_1 = v_{y1} t_d + \frac{1}{2} g t_d^2 \\ x = x_0 + x_q \end{cases} \quad (6)$$

To analyze the success rate of sorting different shapes of gangue with varying qualities, a total of 2250 simulated experiments were conducted using gangue ranging from 70 g to 1500 g. The gangue density in this study was approximately 1800 kg/m^3 . The simulation experiment parameters were set as follows: compressor working pressure of 0.8Mpa, flow rate of $0.97 \text{ m}^3/\text{min}$, nozzle diameter of 8 mm, and belt speed of 0.6 m/s. Figure 7(a–c) shows the sorting results for flat, square and vertical gangue shapes.

After acquiring various types of gangue sorting distance data, we need to determine a reasonable sorting success threshold based on the specific performance parameters of the existing equipment. Through experimental observation, we found that the horizontal displacement limit of coal and gangue particles without pneumatic thrust is 0.26 meters. In view of this, and taking into account the physical spacing of the silos, we set the criterion

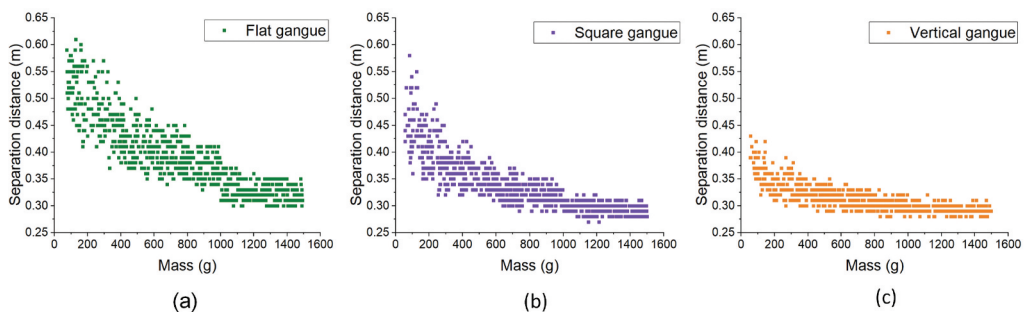


Figure 7. Simulated separation distances for different shapes and masses of gangue, (a) flat (b) square (c) vertical.

for successful sorting as the horizontal movement distance of gangue particles must exceed 0.31 m.

Referring to the statistical data in Figure 7, we take every 100 g as a quality interval, and make detailed statistics on the proportion of successfully sorted gangue in each interval to the total gangue in that interval, and use the results to assess the reliability of different quality and shape of gangue in the sorting process. In addition, we fitted these data into a smooth curve, as shown in Figure 8, for more in-depth analysis.

As can be seen from Figure 8, the flat gangue has the best sorting effect, and its force area is larger. And the same quality of vertical gangue, the force area is smaller, the sorting effect is worse. Square gangue is between these two extremes. The sorting reliability is used for subsequent calculations related to the task allocation algorithm.

Feasibility Analysis of Sorting Task of Manipulator

The manipulator exhibits strong adaptability and flexibility in grasping gangue, allowing it to adjust the jaw opening and closing to successfully grip gangue particles regardless of their shape and size. This characteristic ensures efficient and reliable performance, minimizing the influence of gangue quality and shape on the robot's capabilities. However, during the sorting process, The manipulator needs to complete processes such as tracking, grasping, and placement. Due to the limited capacity of the manipulator to handle a specific number of gangue particles within a given timeframe, task conflicts often arise when multiple tasks are assigned to the manipulator.

As shown in Figure 9, the working areas of the manipulators follow certain manipulator classification rules for ease of calculation:

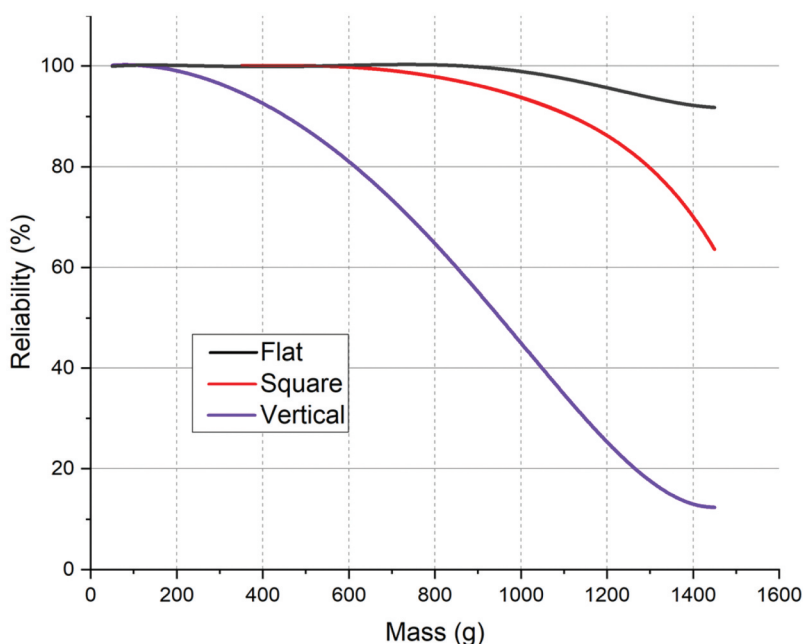


Figure 8. Reliability of gangue sorting for different shapes.

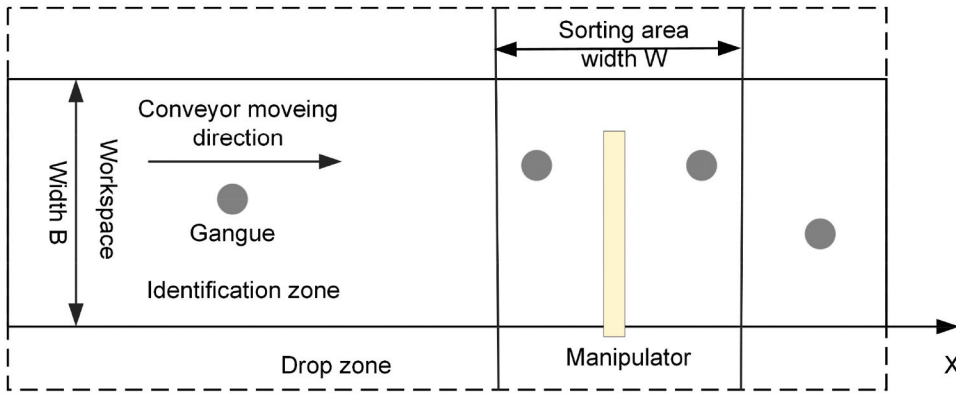


Figure 9. Schematic diagram of the working surface of the robot.

- The ideal plane of the belt conveyor serves as the working plane, maintaining a constant running speed.
- The robot's movement speed is uniform, disregarding acceleration and deceleration processes.
- Manipulator sorting tasks must be completed within the defined work area, and tasks beyond this area are considered unfeasible.
- Communication time delays between the identification module, control system, and execution system of the gangue sorting system are not taken into account.

By calculating the time the manipulator spends grabbing each gangue, as well as the position changes of subsequent gangue during that time, we can assess whether there is a potential conflict between gangue tasks. Ultimately, this analysis enables us to determine the sorting target that the manipulator can effectively accomplish. The calculation process is as follows:

$$G_i : (x_i, y_i, z_i, m_i, s_i) \quad i = 1, 2, \dots, n \quad (7)$$

$$T = \{G_1, G_2, G_3, \dots, G_n\} \quad (8)$$

As shown in Eq. (7) and Eq. (8), G_i represents the i -th gangue individual, where x_i, y_i, z_i represent the coordinate information of the i th gangue particle, m_i represents the quality information of the gangue, s_i represents the shape information of the gangue, and n is the total number of gangue particles, T is the gangue sequence.

Next, we calculate the gripping time of the manipulator. The gripping action of the mechanical arm can be divided into three parts: reset, grasping, and placing. The gripping time of the manipulator can be determined using the Eq. (9), where x_m, y_m, z_m are the coordinates of the position of the jaw, t_1 is the time required for the manipulator to grab the gangue G_i , t_2 is the time required for the manipulator arm to lift and place the gangue into the waste box after grabbing, t_3 is the time required for the manipulator arm to reset. Additionally, C_1 represents the time for the pneumatic claw to open and close, and C_2 represents the time for the manipulator arm to reset, a, b are the coordinates of the position of the waste container. These values are considered constants.

$$\left\{ \begin{array}{l} t_1 = \frac{\sqrt{(x_m - (x_i + v_0 t_1))^2 + (y_m - y_i)^2 + (z_m - \frac{1}{2} z_i)^2}}{v_r} + C_1 \\ t_2 = \frac{z_m - \frac{1}{2} z_i + \sqrt{(x_m - (x_i + v_0 t) - a)^2 + (y_i - b)^2}}{v_r} \\ t_3 = C_2 \\ t = t_1 + t_2 + t_3 \end{array} \right. \quad (9)$$

Then, we determine whether the next gangue particle G_{i+1} can be grasped within the manipulator's workspace. If the condition $x_{i+1} + v_0 t > x_m + w/2$ is satisfied, it indicates that the target gangue cannot be grasped within the manipulator's workspace. We continue this determination process until we find a gangue within the workspace that can be grasped. If the condition $x_m - w/2 < x_{i+2} + v_0 t < x_m + w/2$ is satisfied, it indicates that the manipulator can successfully grasp the G_{i+2} within its workspace. However, if the condition $x_{i+2} + v_0 t < x_m - w/2$ is satisfied, it implies that the G_{i+2} has not reached the manipulator's workspace, resulting in a waiting time as described in Eq. (10). The time used for sorting G_{i+2} targets $t = t_{wait} + t_1 + t_2 + t_3$. By repeating the above steps, we can sequentially determine the sorting targets for the robot while eliminating targets that cannot be sorted.

$$t_{wait} = \frac{x_{i+2} + v_0 t - x_m - \frac{w}{2}}{v_0} \quad (10)$$

Distribution Strategy Based on Genetic Algorithm

The allocation of gangue sorting tasks presents a discrete combinatorial optimization problem, where traditional optimization algorithms struggle to provide sufficiently accurate solutions within a short timeframe. However, it has been demonstrated that heuristic methods, such as genetic algorithms, offer better solutions for this type of problem (Kumar 2020; S. D. Li, Xu, and Zuo 2015; Qi et al. 2022; Sharma et al. 2014; Su et al. 2020). In the collaborative sorting process involving both manipulator and pneumatic separation mechanisms, numerous combinations of gangue particle sizes and shapes exist, along with variations in the reliability of pneumatic separation sorting and the reciprocating movement time of the manipulator. The genetic algorithm, with its ability to explore a wide search space, enables the selection of appropriate parameter settings, coding methods, selection strategies, and customization options to align with the specific optimization requirements. By searching through different combinations, the algorithm identifies the optimal sorting target selection strategy. The algorithm flow is as follows:

(1) Encoding Method

Chromosome encoding is performed using a single chromosome integer encoding method. A sequence of assignments matching the length of the task is generated, which consists of 0, 1. 0 indicates that the task is matched to a pneumatic separation mechanism, and 1 indicates that the task is matched to a manipulator. An example of the coding is shown in Figure 10.

(2) Population Initialization

In order to speed up the algorithm solution, individuals with air-blown sorting reliability above 80% were pre-assigned to pneumatic separation sorting, while those below this threshold were assigned to manipulator sorting. At the same time, to ensure that the range of the solution space is large enough to increase the likelihood of obtaining

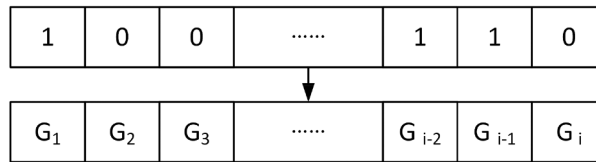


Figure 10. Chromosome coding schematic.

a globally optimal solution, the gene sequences of the remaining 50% of individuals are randomly generated.

(3) Crossover

The algorithm uses task sequence crossover by crossing a random subsequence of the parent individuals and swapping the task order in that subsequence. This helps to introduce new combinations of task sequences to find a better task allocation scheme. The example of the crossover operation is shown in the [Figure 11](#).

(4) Mutation

Mutational manipulation is the alteration of an individual's gene sequence based on the mutation probability P_m . Mutation operations include swap mutation, reverse mutation and point mutation. Swap mutation is the random selection of two genes on a chromosome and swapping them. Reverse mutation involves randomly selecting a subsequence in a chromosome and reversing that subsequence and placing it back in its original position. Point mutation involves selecting a random position in the genome and randomly mutating or replacing the gene values at that position. [Figure 12](#) shows examples of these three types of mutation operations.

(5) Selection

Selection operates by removing less well-adapted individuals from the contemporary population with a certain probability. Through generations of selection, the population becomes genetically better and better. After the crossover operation, the size of the population is doubled, while the selection operation halves the size of the population and

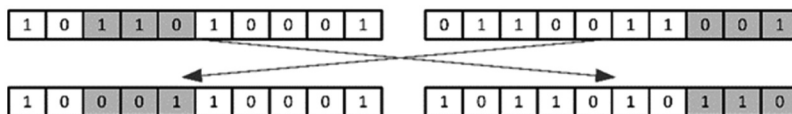


Figure 11. Schematic diagram of mission sequence crossover.

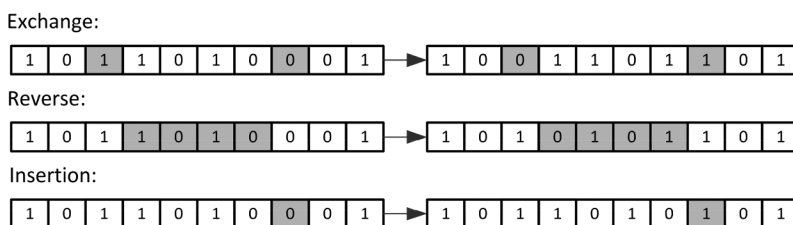


Figure 12. Example of mutation operation.

the population size returns to its original value. In this work, an elitist strategy is used in the selection operation, i.e., the fittest part of the contemporary population is directly retained.

(6) Fitness Function

During the gangue sorting process, each actuator is required to sort only one gangue at a time. The gangue location beyond the sorting area is removed and defined as missed gangue. Uncaptured gangue needs to be added to the pneumatic sorting task list based on location and quality information, and when the reliability of pneumatic sorting of this gangue is less than 10% it is considered as an unsortable target. The fitness is calculated as follows:

For each gangue task list, the sorting benefit is determined using Eq. (11):

$$E = \sum_{i=m}^n p_i \times c_i \times m_i + \sum_{i=p}^q c_i \times m_i \quad (11)$$

Where E represents the total benefit of sorting, p_i is the sorting reliability coefficient calculated by substituting the gangue quality and shape into the reliability curve described above, c_i is the recognition confidence of the i th gangue, given by the gangue recognition model at the time of detection. The first part of the equation represents the benefit of pneumatic separation sorting, while the second part represents the benefit of robotic sorting. To evaluate the task allocation results effectively, the expected sorting benefit (E_v) is calculated using Eq. (12), and the fitness function is determined by the ratio of the actual sorting benefit to the expected benefit, as given in Eq. (13):

$$E_v = \begin{cases} \sum p_i \times c_i \times M_i & p_i \geq 80 \\ \sum c_i \times M_i & p_i < 80 \end{cases} \quad (12)$$

$$fitness = \max\left(\frac{E}{E_v}\right) \quad (13)$$

Materials

Preparation of Experimental Samples

To evaluate the effectiveness of the proposed synergistic sorting method, we compare the sorting performance of single manipulator sorting, pneumatic separation, and the combination of manipulator and pneumatic separation under different belt speeds and gangue content rates. The gangue content rate refers to the percentage of gangue particles larger than 50 mm in the weight of raw coal (Chengliang and Changwu 2003). To ensure accurate calculation of gangue quality and prevent system errors, we weighed the experimental samples and randomly divided them into different gangue content rates, including 4.73%, 9.88%, 16.82%, 21.61%, 25.93%, 30.45%, 35.06%, and 42.16% of the gangue samples, as illustrated in Figure 13.

Experimental Verification

Four sets of tests were set up to verify model performance, In the following, the sorting rate is the ratio of the actual sorting quality to the desired sorting quality, which is used as the evaluation index of the sorting performance.

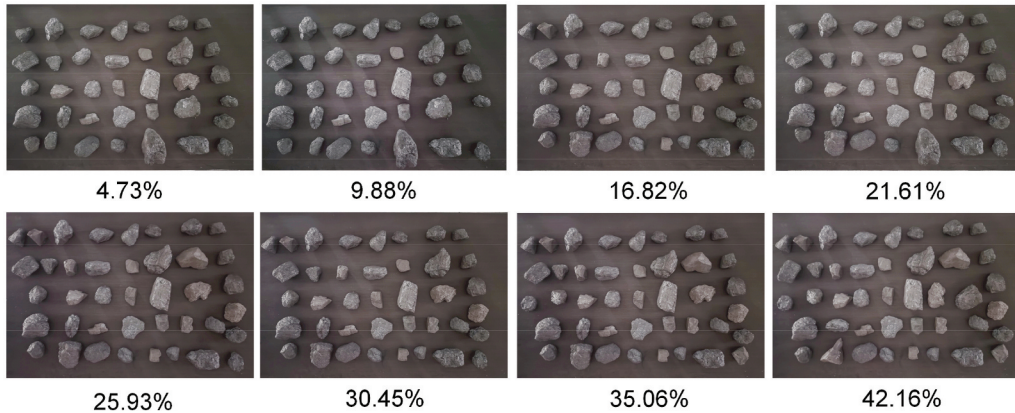


Figure 13. Some samples with different gangue rates.

- (1) Comparison of sorting performance among different sorting methods under the same gangue content and different belt speeds.

A gangue sample with a gangue content of 16.82% was used and the belt speed was set between 0.45 and 0.75 m/s. Comparison of sorting rates for single manipulator, pneumatic separation and collaborative sorting.

- (2) Comparison of sorting performance among different sorting methods with same belt speeds and different gangue content rates.

The belt speed was fixed at 0.6 m/s, and the gangue samples with eight different gangue rates were sorted using the single manipulator sorting, pneumatic separation sorting, and the collaborative sorting method.

- (3) Comparison of sorting rates under different allocation strategies.

With a fixed belt speed of 0.6 m/s, the sorting rates are compared between the First Come First Serve (FCFS) sorting strategy and the allocation strategy optimized using the Genetic Algorithm (GA).

- (4) GA algorithm stability analysis.

The coal gangue collaborative sorting task allocation problem is similar to the Traveling Salesman Problem (TSP) in that they are both NP-Hard (Non-deterministic Polynomial) problems. During the coal gangue sorting process, the gangue is in continuous motion, and we must make the best possible allocation scheme within a limited time. The allowed runtime of the program depends on the buffer distance between the recognition area and the sorting mechanism, the belt speed, and the performance of the upper computer. The buffer distance of the experimental platform is approximately 2 m, the belt speed is set to 0.6 m/s, and the upper computer processor is an Intel Core™ i5-7300HQ. The maximum runtime of the program is set to 3s. To avoid randomness, 50 simulation experiments are conducted for each gangue content rate, and the stability of the algorithm is evaluated by comparing the best, worst, and average values.

Results and Discussion

(1) Performance among different sorting methods under the same gangue content and different belt speeds.

Figure 14 illustrates the sorting rates achieved by the three sorting methods. It can be observed that, for the same set of experimental samples under different belt speeds, Collaborative sorting methods combining pneumatic separation and robotic sorting had sorting rates 28.69% to 29.31% higher than methods using pneumatic separation sorting alone. Compared to single manipulator sorting, the sorting rate is improved by 54.08% to 69.68%. In the absence of communication delays, the pneumatic separation sorting method exhibits a millisecond-level response time, which is much shorter than the increased frequency of gangue particle descent caused by the higher belt speed. Therefore, the impact on pneumatic separation sorting is minimal. However, for single manipulator sorting, as the belt speed increases, the time for gangue particles to pass through the sorting area decreases from 2.2s to 1.3s. Some gangue particles may enter and exceed the sorting area during the execution of sorting actions by the manipulator, resulting in an increased number of missed detections and a gradual decline in the sorting rate. Moreover, a higher belt speed leads to a larger number of missed detections. Under different belt speeds, the collaborative sorting strategy adjusts and optimizes the targets for the manipulator and pneumatic separation sorting, determining the sorting scheme that maximizes sorting benefits for each belt speed, thus adapting to changes in belt speed.

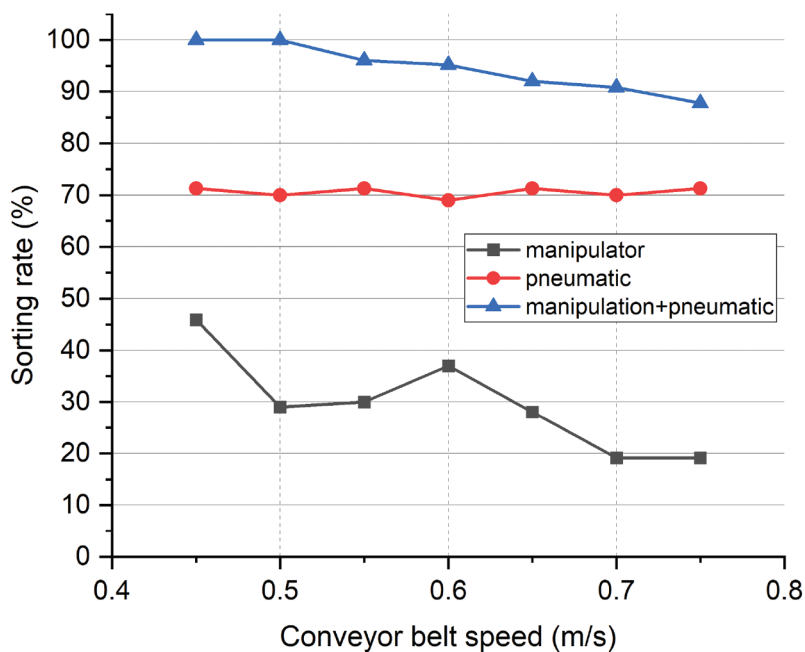


Figure 14. Comparison of sorting rates of three sorting methods at different belt speeds.

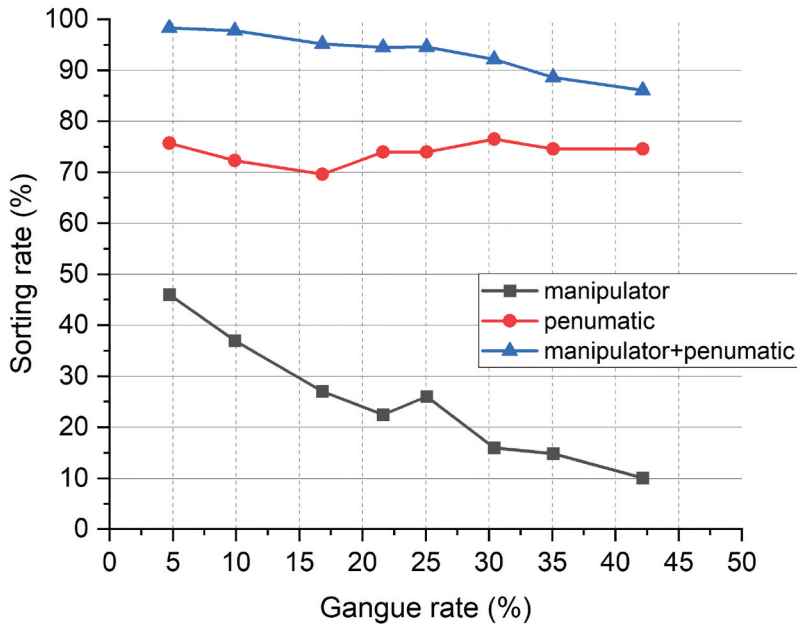


Figure 15. Comparison of sorting rates of three sorting methods with different gangue rates.

(2) Performance of different sorting methods at the same belt speed and different gangue rates.

From Figure 15, it can be observed that the proposed synergistic sorting method outperforms the other two sorting methods across different gangue content rates. The sorting rate of the synergistic sorting method is approximately 20.66% higher than that of the pneumatic separation sorting on average. As the gangue content rate increases, the response frequency of the pneumatic separation mechanism per unit time increases. However, the impact of gangue content rate increase is relatively small as long as it has not reached the limit speed of the electromagnetic valve response.

The time required for a single manipulator to complete a sorting action depends on the position of the gangue. As the gangue content rate increased, the density of gangue on the conveyor belt increased as well. Consequently, when the manipulator sorted a specific gangue particle, other nearby gangue particles might surpass the sorting area during the placement process, resulting in missed detections. Thus, the sorting rate of the manipulator gradually decreased with higher gangue content rates. In fact, the performance difference between the manipulator sorting and the synergistic sorting method could reach up to 76.09%.

The increase of gangue density on the conveyor belt, the number of gangue on different distance segments will also increase, in the process of the iteration of the genetic algorithm, the manipulator in the near distance in the low reliability of the gangue below the grabbing low reliability gangue and the grabbing of the global low reliability gangue to achieve a balance, in other words, to achieve a balance between the frequency of the grasp and the quality of the grasp, the purpose is to improve the overall efficiency of the sorting and to maintain the stability of the performance. Even so, sorting rates still showed a decline, mainly due to the limited sorting speed of individual manipulators.

(3) Comparison of sorting rates under FCFS and GA task Assignment Strategies.

In order to try to avoid a large number of missed sorting or task conflicts caused by unreasonable threshold settings, the sorting threshold of the manipulator in the FCFS sorting strategy was set to 1300 g through field tests, which means that gangue weighing more than 1300 g will be assigned as the sorting target of the manipulator and the rest will be sorted by the air blower. If task conflicts arise, the current task is prioritized, while the remaining gangue particles are assigned as targets for pneumatic separation sorting. The experimental results are shown in Figure 16.

From Figure 15, it can be observed that as the gangue content rate increases, the picking rate decreases significantly when using the FCFS allocation strategy. This is because the FCFS strategy prioritizes the manipulator to sort gangue particles with higher mass, while higher mass gangue particles do not necessarily exhibit lower reliability in pneumatic separation sorting. For example, a flat gangue particle weighing 1.5 kg still maintains a 90% reliability for pneumatic separation sorting.

In addition, the position of gangue particles on the conveyor belt has a great influence on the sorting time of the manipulator. FCFS triggers the sorting action according to a set threshold. When sorting a gangue that is far away from the gangue, it may miss the higher-value target that is similar to the gangue in terms of distance, but with a lagging position. With the increase of the density of gangue particles on the conveyor belt, such a disadvantage of the FCFS allocation strategy becomes even more obvious. In contrast, the GA-based allocation strategy is based on a certain length of the gangue particle sequence for global adjustment of the allocation scheme, avoiding the above situation and providing an optimal allocation scheme. The GA-based allocation strategy has more obvious improvement on the sorting efficiency, especially in the case of high gangue content.

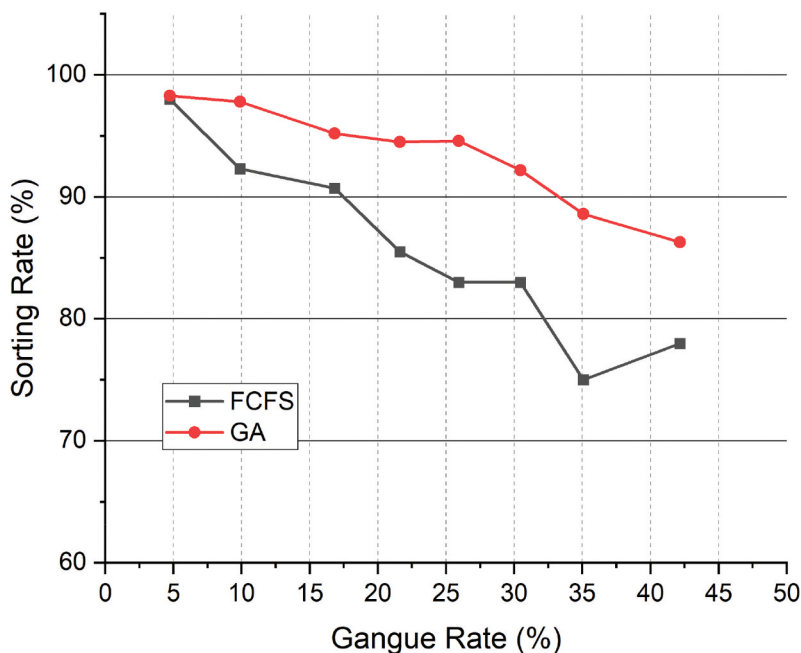


Figure 16. Comparison of sorting rates under FCFS and GA task assignment strategies.

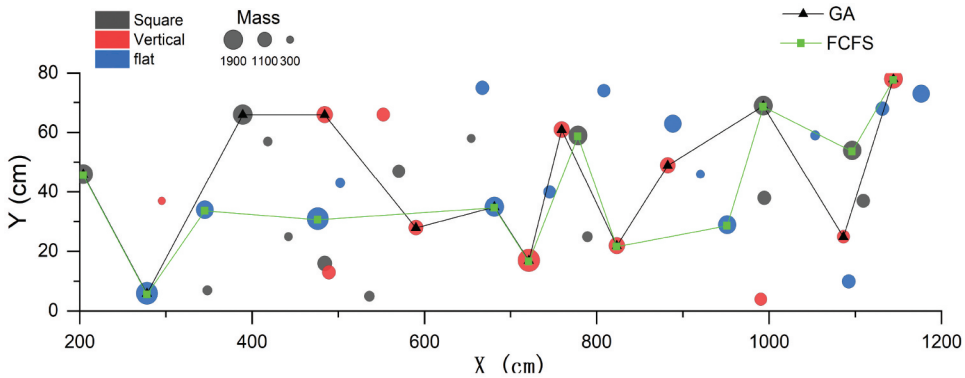


Figure 17. Comparison of gripping order of manipulators under FCFS and GA task assignment strategies with 16.8% gangue rate.

Figure 17 shows the path comparison between the FCFS allocation strategy and the GA-based allocation strategy when sorting the same gangue sequence with a gangue content of 16.82%. The green line represents the FCFS strategy, and the black line represents the GA-based allocation strategy, and the solid lines in the figure are not the actual motion paths of the manipulator, but just indicate the order of the grasping tasks.

According to the different gripping paths produced by the two task allocation strategies, it can be seen that the genetic algorithm-based task allocation strategy can make the manipulator action more reasonable compared with the FCFS strategy with fixed mass threshold. It can flexibly select the gangue with large volume but low reliability of air blow sorting and the gangue with medium to high mass but low reliability of air blowing mechanism. This flexibility is manifested in the proportion of sorted gangue quality achieved by the manipulator. As shown in Figure 18, for flat-shaped gangue, the proportion of pneumatic sorting quality is consistently higher than that of the manipulator. However, for square-shaped ores, the proportion of manipulator sorting quality remains above 60%, and for vertical-shaped gangue, the proportion of manipulator sorting quality reaches as high as 63% to 100%. This reflects the priority of the allocation strategy for sorting ores of different shapes, namely giving priority to the manipulator for sorting ores with low reliability in pneumatic sorting.

(4) Stability performance of GA-based task allocation algorithms

The stability of the algorithm, as shown in Figure 19, is assessed based on the fluctuation of sorting efficiency between the optimal and worst-case scenarios, as well as the average performance. Due to the inability to exhaustively explore all possibilities and find the optimal solution within a limited time for NP-Hard problems, the fluctuation of sorting efficiency in a single task allocation process is below 3.2% for the optimal solution, with a minimum fluctuation of about 1.2%. This indicates that the algorithm remains relatively stable in practical usage.

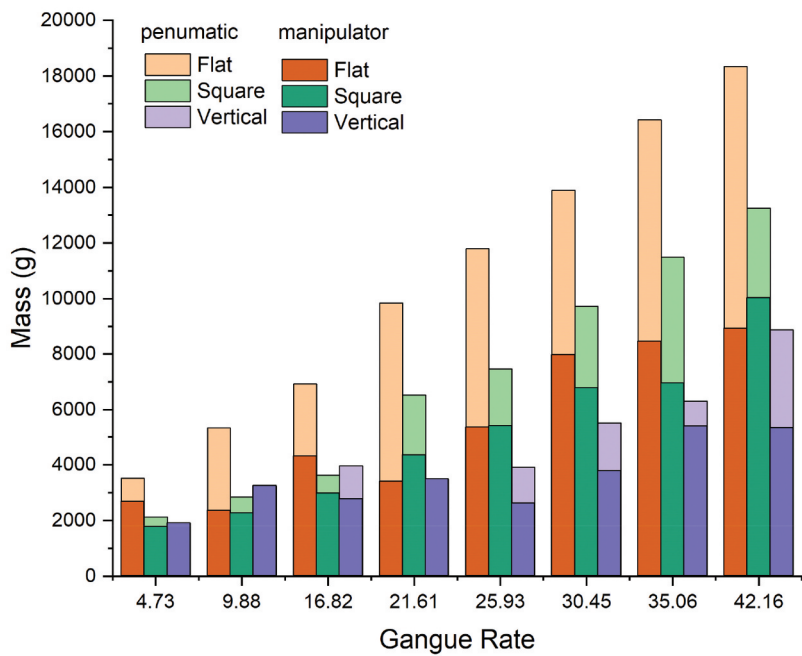


Figure 18. Mass distribution of gangue sorted by air blowing and manipulator under different gangue content rates.

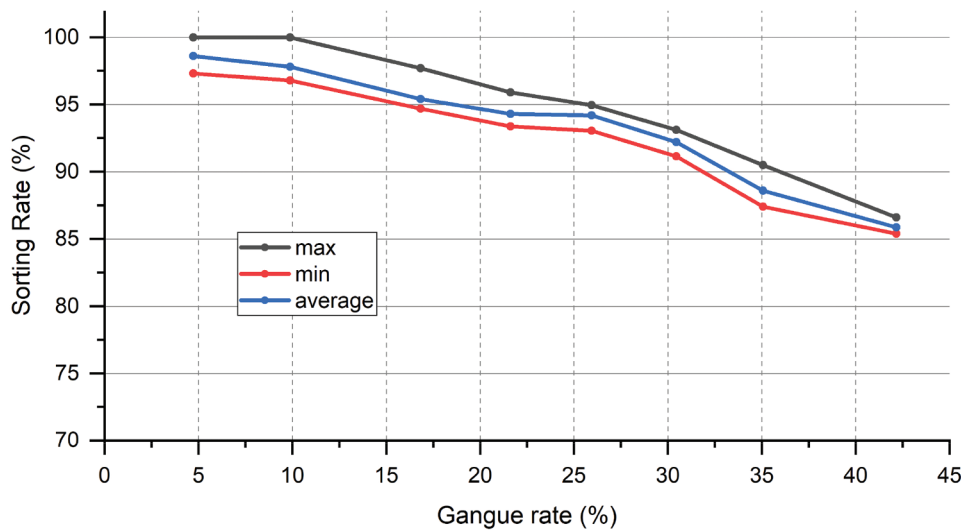


Figure 19. Under different gangue rates, the highest value, the lowest value and the average value of the gangue picking rate of the program obtained by solving 50 times.

Conclusion

Coal gangue automatic sorting equipment is an important part of coal gangue intelligent sorting system. Aiming at the limitations of the existing gangue sorting technology, we propose a gangue sorting method with pneumatic separation and robotic synergy. The air blowing and the manipulator work together through the task allocation strategy based on genetic algorithm. Experimental results prove that our collaborative sorting method for coal gangue increases sorting efficiency by an average of 20.66% compared to pneumatic separation across various belt speeds and gangue contents. Compared to single manipulator sorting, collaborative sorting boosts efficiency by up to 76.09%, enabling efficient coal gangue sorting. The cooperative sorting method can be reasonably expanded to include pneumatic sorting and multiple manipulators, based on actual demand, to fulfill specific sorting needs cost-effectively.

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References

- Ambrós, W. M. 2020. "Jigging: A Review of Fundamentals and Future Directions." *Minerals* 10 (11): 998. <https://doi.org/10.3390/min10110998>.
- Bi, H. J., H. B. Zhu, L. Zu, S. H. He, Y. Gao, and S. Gao. 2019. "Pneumatic Separation and Recycling of Anode and Cathode Materials from Spent Lithium Iron Phosphate Batteries." *Waste Management & Research* 37 (4): 374–385. <https://doi.org/10.1177/0734242x18823939>.
- Chengliang, Y., and C. Changwu. 2003. "Development of National Standards for Coal Gangue Ratio and Its Determination Methods." *China Standards Review* 2003 (10):3.
- De Almeida, A., S. Nicolas, O. P. Taranto, and M. A. Silva. 2019. "Theoretical Study of Pneumatic Separation of Sugarcane Bagasse Particles." *Biomass & Bioenergy* 127:105256. <https://doi.org/10.1016/j.biombioe.2019.105256>.
- Fan, G. W., D. S. Zhang, and X. F. Wang. 2014. "Reduction and Utilization of Coal Mine Waste Rock in China: A Case Study in Tiefa Coalfield." *Resources, Conservation and Recycling* 83:24–33. <https://doi.org/10.1016/j.resconrec.2013.12.001>.
- Jiang, J. H., Y. F. Han, H. J. Zhao, J. L. Suo, and Q. B. Cao. 2023. "Recognition and Sorting of Coal and Gangue Based on Image Process and Multilayer Perceptron." *International Journal of Coal Preparation and Utilization* 43 (1): 54–72. <https://doi.org/10.1080/19392699.2021.2002852>.
- Kumar, A. 2020. "Improved Genetic Algorithm to Solve Small Scale Travelling Salesman Problem." International Conference on Intelligent Computing and Control Systems (ICICCS), 516–520. Madurai, INDIA. Anna University, Vaigai College Engineering.
- Li, J., T. Dong, and Y. Li. 2016. "Research on Task Allocation in Multiple Logistics Robots Based on an Improved Ant Colony Algorithm." 2016 International Conference on Robotics and Automation Engineering (ICRAE), 17–20, Jeju, Korea. <https://doi.org/10.1109/ICRAE.2016.7738780>

- Li, S. D., X. Xu, and L. Zuo. 2015. "Task Assignment of Multi-Robot Systems Based on Improved Genetic Algorithms." In IEEE International Conference on Mechatronics & Automation, 1430–1435. Beijing, China. People's Republic of China.
- Liu, P., H. W. Ma, X. H. Zhang, and X. Z. Qiao. 2019. "On the Equivalent Position Workspace for a Coal Gangue Picking Robot." In 3rd International Conference on Artificial Intelligence, Automation and Control Technologies (AIAC), 25–27, Vol. 1267, Xian. People's Republic of China.
- Lv, Z. Q., W. D. Wang, K. H. Zhang, W. J. Li, J. D. Feng, and Z. Q. Xu. 2022. "A Synchronous Detection-Segmentation Method for Oversized Gangue on a Coal Preparation Plant Based on Multi-Task Learning." *Minerals Engineering* 187:107806. <https://doi.org/10.1016/j.mineng.2022.107806>.
- Meyer, E. J., and I. K. Craig. 2015. "Dynamic Model for a Dense Medium Drum Separator in Coal Beneficiation." *Minerals Engineering* 77:78–85. <https://doi.org/10.1016/j.mineng.2015.03.002>.
- Qi, B., L. Pu, C. Xu, and A. Zheng. 2022. "Multi-Robot Task Assignment Method in the Construction Waste Sorting System." 2022 IEEE International Conference on Mechatronics and Automation (ICMA), 1364–1369, Guilin, Guangxi, China. <https://doi.org/10.1109/ICMA54519.2022.9856058>
- Sariel, S., and T. Balch. 2006. "A Distributed Multi-Robot Cooperation Framework for Real Time Task Achievement." In 8th International Symposium on Distributed Autonomous Robotic Systems (DARS 8), 187–196. Minneapolis, MN. University Minnesota.
- Shang, D. Y., Y. W. Wang, Z. Y. Yang, J. J. Wang, and Y. Liu. 2020. "Study on Comprehensive Calibration and Image Sieving for Coal-Gangue Separation Parallel Robot." *Applied Sciences-Basel* 10 (20): 7059. <https://doi.org/10.3390/app10207059>.
- Sharma, A., N. Singh, A. Hans, and K. Kumar. 2014. "Review of Task Scheduling Algorithms Using Genetic Approach." In International Conference on Innovative Applications of Computational Intelligence on Power Energy and Controls with their Impact on Humanity (CIPECH), 169–172. Ghaziabad, INDIA. Krishna Institute of Technology.
- Su, J. R., J. T. Qi, C. Wu, M. M. Wang, and J. J. Guo. 2020. "Multi-UAVs Target Attack Based on Improved Genetic Algorithm." In 39th Chinese Control Conference (CCC), 1466–1471. Shenyang. People's Republic of China.
- Sun, Z. Y., L. L. Huang, and R. Q. Jia. 2021. "Coal and Gangue Separating Robot System Based on Computer Vision." *Sensors* 21 (4): 1349. <https://doi.org/10.3390/s21041349>.
- Wang, P., H. W. Ma, Y. Zhang, X. G. Cao, X. D. Wu, H. L. Qiao, and J. Xia. 2023. "A Cooperative Strategy of Multi-Arm Coal Gangue Sorting Robot Based on Immune Dynamic Workspace." *International Journal of Coal Preparation and Utilization* 43 (5): 794–814. <https://doi.org/10.1080/19392699.2022.2078808>.
- Xiangang, C., F. Jiahao, W. Peng, L. I. Ning, and S. U. Lingling. 2019. "Study on Coal-Gangue Sorting Method Based on Multi-Manipulator Collaboration." *Coal Science and Technology* 47:7–12. <https://doi.org/10.3390/s22207987>.
- Xin, P., Z. X. Dong. 2022. "Intelligent Coal Gangue Sorting System Based on YOLOv5." Asia Conference on Algorithms, Computing and Machine Learning (CACML), 482–486, Hangzhou, China: IEEE. <https://doi.org/10.1109/CACML55074.2022.00088>
- Yang, Y., and Q. Zeng. 2021. "Multipoint Acceleration Information Acquisition of the Impact Experiments Between Coal Gangue and the Metal Plate and Coal Gangue Recognition Based on SVM and Serial Splicing Data." *Arabian Journal for Science and Engineering* 46 (3): 2749–2768. <https://doi.org/10.1007/s13369-020-05227-6>.
- Yang, Y., Q. L. Zeng, G. J. Yin, and L. R. Wan. 2019. "Vibration Test of Single Coal Gangue Particle Directly Impacting the Metal Plate and the Study of Coal Gangue Recognition Based on Vibration Signal and Packing Integration." *Institute of Electrical and Electronics Engineers Access* 7:106783–106804. <https://doi.org/10.1109/access.2019.2932118>.
- Zhao, Y., X. Yang, Z. Luo, C. Duan, and S. Song. 2014. "Progress in Developments of Dry Coal Beneficiation." *International Journal of Coal Science & Technology* 1 (1): 103–112. <https://doi.org/10.1007/S40789-014-0014-5>.

- Zheng, K. H., C. L. Du, J. P. Li, and B. J. Qiu. 2016. "Coal and Gangue Underground Pneumatic Separation Effect Evaluation Influenced by Different Airflow Directions." *Advances in Materials Science and Engineering* 2016:1–13. <https://doi.org/10.1155/2016/6465983>.
- Zheng, K. H., C. L. Du, J. P. Li, B. J. Qiu, and D. L. Yang. 2015. "Underground Pneumatic Separation of Coal and Gangue with Large Size (≥ 50 Mm) in Green Mining Based on the Machine Vision System." *Powder Technology* 278:223–233. <https://doi.org/10.1016/j.powtec.2015.03.027>.
- Zhou, Y., B. Albijanic, B. Tadesse, Y. Wang, J. Yang, and X. Zhu. 2020. "Surface Properties of Aged Coal and Their Effects on Bubble–Particle Attachment During Flotation – ScienceDirect." *Advanced Powder Technology* 31 (4): 1490–1499. <https://doi.org/10.1016/j.appt.2020.01.029>.
- Zou, L., X. Yu, M. Li, M. Lei, and H. Yu. 2020. "Nondestructive Identification of Coal and Gangue via Near-Infrared Spectroscopy Based on Improved Broad Learning." *IEEE Transactions on Instrumentation and Measurement* 69 (10): 8043–8052. <https://doi.org/10.1109/tim.2020.2988169>.